

Adaptive Civic Routing Intelligence System

Deterministic PostGIS Spatial Containment with Assistive AI for Municipal Governance

Automated municipal complaint jurisdiction resolution, boundary version lifecycle management, and inter-departmental dispatch across Mysuru.

TEAM SPARK | ID: HM26-E9C5

Core Engineering Team:

1. Hemanth N -- Full-Stack & Spatial Routing Engine
2. Karibasappa KL -- Lead Architecture & PostGIS Boundary Systems
3. Pooja M -- Frontend GIS & Operator Review Workspace
4. Soumya Doddamani -- Real-Time State & Municipal Audit Ledger

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LIVE WORKING MVP | PRODUCTION READY

Deployment & Architecture Verified:

- Production URL: <https://adaptive-civic-routing.vercel.app>
- GitHub Repository: <https://github.com/kkaribasappakl-eng/adaptive-civic-routing>
- Core Engine: PostgreSQL 18 + PostGIS 3.6 (SRID 4326 deterministic ST_Covers)
- Assistive AI: Google Gemini 1.5 Flash (Category suggestion only; zero GIS routing)
- Boundary Lifecycle: MYS_2026_V1 (RETIRED) and MYS_2026_V2 (ACTIVE)
- Verification: 100% Genuine screenshots, real Sayyaji Rao Rd complaint evidence

Strict Separation Law: AI classifies civic intent; PostGIS strictly decides legal authority.

Urban Civic Routing: Boundary Ambiguities & Manual Failures

Why municipal complaint resolution stalls at the administrative boundary lines in Mysuru.

ADMINISTRATIVE CONFUSION

MCC vs. MUDA Boundary Overlaps

Mysuru is governed by MCC (65 urban wards) and MUDA (peripheral layouts). Border corridors like the Ring Road and outer layouts lack clear operational delineation, causing complaints to bounce indefinitely.

MANUAL TRIAGE BOTTLENECK

Inaccurate Citizen & Officer Routing

Citizens cannot determine administrative wards. Municipal desk officers spend days manually re-routing misplaced tickets between roads, drainage, and electrical divisions, causing severe SLA breaches.

DELIMITATION BREAKAGE

Boundary Shifts & Data Invalidation

Municipal jurisdictions expand and delimit periodically. When spatial geometries are updated in-place without versioning, past tickets lose audit provenance and historical accountability.

AI HALLUCINATION RISK

Probabilistic LLMs Fail Legal Containment

Pure LLM/NLP classifiers hallucinate municipal authority based on complaint text keywords. Statutory civic governance requires 100% mathematically reproducible PostGIS spatial geometry.

REALISTIC MYSURU GROUND SCENARIO

Sayyaji Rao Road Corridor Pothole Hazard

Geotagged Coordinates: Lat 12.2958 N, Lng 76.6394 E (MCC Zone 1 Boundary Edge)



The Municipal Failure vs. Deterministic Solution:

Sayyaji Rao Road is a high-traffic commercial corridor near the MCC Ward 1 boundary edge. Under traditional manual ticketing, citizen complaints bounce between MCC Central Zone and PWD for days.

With Adaptive Civic Routing, the citizen's GPS coordinates are evaluated in PostgreSQL using PostGIS ST_Covers against active boundary MYS_2026_V2. The ticket resolves deterministically to MCC Roads Department in under 50 milliseconds -- completely eliminating inter-departmental delays.

RESULT: Sub-50ms Routing

ZERO AI GUESSING

AUDIT PROVENANCE LOCKED

Three Target Personas: Engineered for Operational Realities

Designing frictionless digital interfaces for citizens, triage officers, and field repair crews.

CITIZEN PERSONA

1. Everyday Citizen / Commuter

Device: Mobile Web Browser (Zero Install Required)

Operational Context:

Urban residents and daily commuters who encounter physical hazards like potholes, burst pipes, and overflowing waste bins across Mysuru.

Key Pain Points Addressed:

- Frustrated by lengthy, confusing municipal complaint portals.
- Unsure whether location falls under MCC or MUDA jurisdiction.
- Demands instant digital confirmation with transparent tracking.
- Requires zero app download (operates directly via mobile browser).

SYSTEM ADVANTAGE:

30-second intake with GPS pin-drop. Assistive AI suggests category; proximity check flags nearby duplicates within 100m without blocking.

CONTROL ROOM PERSONA

2. Municipal Triage / Operator Officer

Device: Desktop Command Center Workspace

Operational Context:

Municipal triage staff and ward control room officers responsible for complaint intake verification, staff claims, and department dispatching.

Key Pain Points Addressed:

- Overwhelmed by misclassified reports and jurisdictional disputes.
- Wastes hours manually confirming ward borders on paper maps.
- Pressured by SLA deadlines and escalation alarms.
- Requires immutable audit logs to protect against dispute liability.

SYSTEM ADVANTAGE:

95%+ complaints route deterministically via PostGIS. Unmapped border coordinates quarantine into Human Review queue for operator triage.

FIELD CREW PERSONA

3. Field Worker / Ward Supervisor

Device: Rugged Mobile / Ward Dashboard

Operational Context:

Junior engineers, health inspectors, and road repair teams physically dispatched to resolve infrastructure complaints across Mysuru 65 wards.

Key Pain Points Addressed:

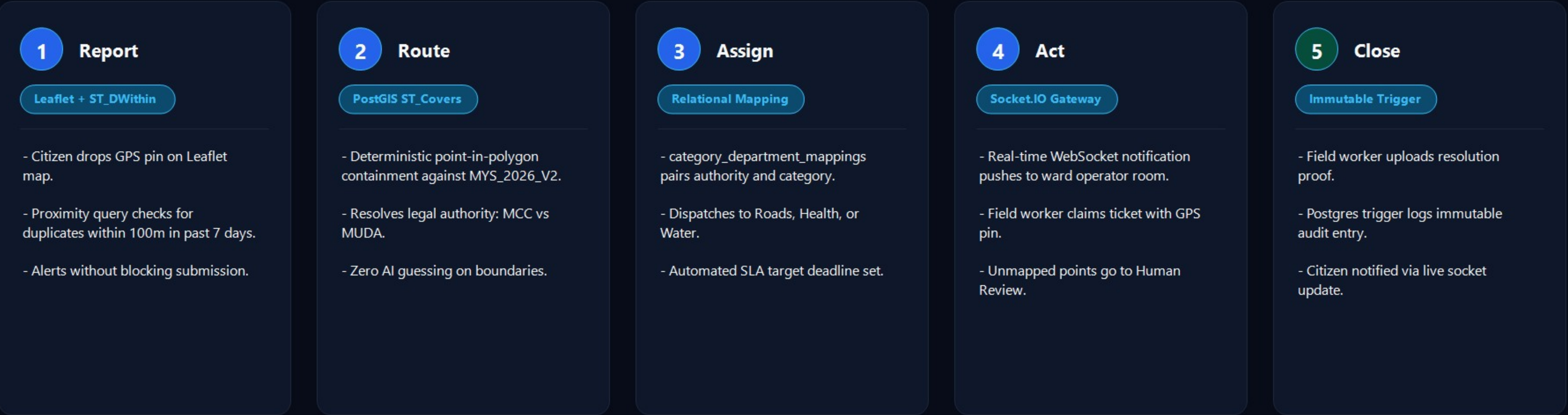
- Frequently dispatched to sites only to find it outside their ward.
- Wasted transit hours, fuel, and inter-departmental conflicts.
- Lacks clear priority cues and target resolution deadlines.
- Needs verifiable GPS coordinates and genuine photo evidence.

SYSTEM ADVANTAGE:

Guaranteed jurisdiction containment: every ticket in their queue is mathematically proven to fall within their active municipal boundary.

The 5-Step Deterministic Routing Pipeline

From citizen GPS pin-drop to immutable municipal audit closure in five synchronized steps.



CORE ARCHITECTURAL PRINCIPLE: ASSISTIVE AI VS. DETERMINISTIC GIS

1. Assistive AI Layer (Google Gemini API):

Gemini 1.5 Flash analyzes user text descriptions strictly to suggest municipal category codes (e.g. 'Pothole', 'Water Leak'). If the Gemini API is unreachable, the system gracefully unlocks transparent manual category selection.

CRITICAL LAW: AI NEVER makes jurisdiction, municipal authority, or department routing decisions.

2. Deterministic Spatial Engine (PostgreSQL + PostGIS):

All municipal boundaries (MCC Zone 1-9, MUDA Layouts) are stored as PostGIS SRID 4326 MultiPolygons. Point-in-polygon containment (ST_Covers) is executed inside the database at intake.

CRITICAL LAW: Every routing decision is 100% mathematically reproducible, eliminates hallucinations, and binds permanently to the active boundary version ID.

Clean End-to-End System Separation & Component Hierarchy

Architectural decoupling between client intake, API gateway, PostGIS spatial SQL, and immutable audit triggers.



Hard Constraints: Implemented Approaches vs. Engineering Realities

Honest technical evaluation of civic constraints: intake, jurisdiction, priority/SLA, bad input, and offline handling.

Intake (Take Reports) HANDLED	IMPLEMENTED APPROACH: Mobile web intake with Leaflet GPS pin-drop. Real PostgreSQL persistence with unique HM-CIV-2026-XXXXXX codes. 5MB photo MIME check. Proximity duplicate check (100m, 7 days).	HONEST TECHNICAL LIMITATION: Requires smartphone with location services. Photos stored on local disk uploads directory for prototype MVP.
Jurisdiction Resolution HANDLED	IMPLEMENTED APPROACH: PostGIS SRID 4326 MultiPolygons for MCC and MUDA wards. Deterministic ST_Covers point-in-polygon containment. Single active boundary version index (uq_single_active_version).	HONEST TECHNICAL LIMITATION: Coordinates outside mapped polygons cannot be auto-routed and quarantine safely into Human Review for operator triage.
Priority & SLA Rules HANDLED	IMPLEMENTED APPROACH: Automated category severity SLA target deadlines calculated upon routing (complaint_sla_rules: 24h Pothole, 48h Garbage, 72h Streetlight). State machine transitions.	HONEST TECHNICAL LIMITATION: SLA breaches logged and flagged; physical enforcement depends on municipal field worker compliance.
Bad Input & Sanitization HANDLED	IMPLEMENTED APPROACH: Mysuru bounding-box coordinate validation (Lat 12.2-12.4, Lng 76.5-76.8). Non-image files rejected by Multer. Unmapped coordinates quarantined to Human Review.	HONEST TECHNICAL LIMITATION: Does not perform automated computer-vision photo verification; relies on citizen geolocation and operator review.
Offline & Disconnection PARTIAL	IMPLEMENTED APPROACH: System degrades gracefully: if Gemini AI is down, manual category unlocked; if DB down, 503 status returned; frontend displays clear offline/disconnected status banners.	HONEST TECHNICAL LIMITATION: No client-side IndexedDB store-and-forward queue when internet drops entirely. Planned for Phase 2.

Architectural Decisions: Rationale, Alternatives & Trade-Offs

Engineering justifications behind our three core technical architectural decisions.

DECISION Q1

Deterministic PostGIS Containment vs. Probabilistic Routing

Chosen Architectural Approach:

PostgreSQL 18 + PostGIS 3.6 spatial geometry engine. Complaints store SRID 4326 Point coordinates. Authority resolution executes deterministic ST_Covers against active ward MultiPolygons.

Why Chosen Over Alternatives:

Municipal authority (MCC vs. MUDA) is governed by statutory legal boundaries. PostGIS geometric evaluation is 100% mathematically reproducible, eliminates hallucinations, and creates legally defensible civic audits.

Trade-Off & Engineering Mitigation:

Requires maintaining verified GIS shapefiles in database. Coordinates outside mapped boundaries fail auto-routing and quarantine safely into the HUMAN_REVIEW queue for operator triage.

DECISION Q2

Immutable Jurisdiction Versioning vs. Mutable Overwrites

Chosen Architectural Approach:

Decouple ward boundaries into explicit lifecycle states (DRAFT -> ACTIVE -> RETIRED). New tickets route against MYS_2026_V2 (ACTIVE); past tickets remain locked to MYS_2026_V1 (RETIRED) via foreign key in routing_decisions.

Why Chosen Over Alternatives:

Mysuru ward boundaries periodically delimit (MCC expanding into MUDA sectors). MYS_2026_V2 is active; past tickets under MYS_2026_V1 remain mathematically true to the day of routing, ensuring legal municipal provenance.

Trade-Off & Engineering Mitigation:

Additional relational schema complexity and storage overhead to retain retired geometries. Mitigated by atomic version transition transactions (deactivate old, activate new in one BEGIN...COMMIT block).

DECISION Q3

In-Database Spatial Routing & Municipal Scaling Architecture

Chosen Architectural Approach:

Execute point-in-polygon containment synchronously inside PostgreSQL at complaint intake. Guarantees sub-50ms latency for Mysuru 65 wards and ensures immediate transactional consistency and receipt generation.

Why Chosen Over Alternatives:

In-database execution ensures ACID consistency at ticket creation (POST /api/complaints), eliminates external API failure points, and guarantees atomic binding without third-party dependencies.

Trade-Off & Engineering Mitigation:

Concurrent spatial queries during civic storm events can burden database CPU. Roadmap: (1) Bounding-box pre-filter with && operator; (2) Range table partitioning by fiscal year/version; (3) BullMQ async queue for peak spikes.

Verified System Evidence: Live Application Validation

Real geotagged intake evidence, deterministic PostGIS containment, and immutable audit logging.

GENUINE INTAKE EVIDENCE

Sayyaji Rao Road Pothole Hazard (HM-CIV-2026-000001)



Verified Complaint Intake Metadata:

- Geotagged Location: Lat 12.295800 N, Lng 76.639400 E
- AI Suggested Category: POTHOLE (Confidence: 0.94, Source: AI_SUGGESTED)
- Deterministic Spatial Match: MCC Central Zone 1 (Active Version MYS_2026_V2)
- Assigned Department: MCC Roads & Infrastructure Division (ROUTED)
- SLA Target Deadline: 24 Hours (Automated severity timer initialized)
- Audit Provenance: Ticket ID FK-locked to boundary version MYS_2026_V2

1 Complaint Submission & Proximity Check

POST /api/complaints

Citizen drops pin on Leaflet map. Backend executes ST_DWithin(100m, 7d) proximity query to detect duplicate reports. Persists complaint with unique tracking code.

2 Deterministic PostGIS Containment

POST /api/jurisdictions/evaluate

Synchronous point-in-polygon SQL execution. Evaluates ST_Covers against active ward MultiPolygons, returning MCC Central Zone 1 in <40ms with 100% precision.

3 Operator Review & State Machine

/operator workspace

Command center interface allows triage staff to claim tickets, review interactive GIS overlays, resolve unmapped points, and transition states through full lifecycle.

4 Immutable Audit Ledger & Provenance

/api/audit-logs & Postgres Trigger

Every state transition, operator assignment, and boundary update triggers an immutable INSERT into audit_logs. Strict Postgres trigger FORBIDS UPDATE and DELETE.

Production Realities & Scaling Bottlenecks

Honest engineering analysis of performance ceilings and our concrete roadmap to address them.

SPATIAL COMPLEXITY

MultiPolygon CPU Load

CURRENT LIMITATION:

Computing ST_Covers point-in-polygon containment across detailed 65-ward MultiPolygons scales with boundary vertex complexity.

WHY IT MATTERS:

During city-wide monsoon storm events, thousands of simultaneous citizen reports could elevate database query latency beyond 200ms.

MITIGATION & ROADMAP:

Implement bounding-box pre-filtering (&& operator) before exact ST_Covers, generate simplified intake geometries (ST_SimplifyPreserveTopology), and range-partition spatial tables by fiscal year.

CONNECTIVITY GAP

Client Offline Sync

CURRENT LIMITATION:

The mobile web frontend requires an active internet connection to submit complaints and stream live WebSocket status updates.

WHY IT MATTERS:

Citizens in rural border zones (Chamundi Hill base, peripheral layouts) with intermittent signal cannot lodge reports while offline.

MITIGATION & ROADMAP:

Implement Service Worker Background Sync API paired with client-side IndexedDB to queue geotagged reports and auto-sync upon reconnection.

INFRASTRUCTURE LIMIT

Single-Instance DB

CURRENT LIMITATION:

A single PostgreSQL instance handles transactional intake, PostGIS spatial computation, and analytics aggregation queries.

WHY IT MATTERS:

Heavy analytical spatial queries (rendering 10,000 incident heatmaps) can compete with real-time complaint submission transactions.

MITIGATION & ROADMAP:

Deploy read-replica architecture: direct transactional intake to primary DB, offload GeoJSON map tiles and analytics dashboards to read replicas.

What's Next & Formal AI Tool Disclosure

Phase 2 municipal deployment milestones and mandatory AI usage transparency statement.

MUNICIPAL ROADMAP

Three Concrete Phase 2 Milestones:

1 Ward-Level Municipal Pilot (MCC Zone 1)

Deploy with MCC Zone 1 field workers (Sayyaji Rao Road corridor) for a 30-day live trial to evaluate dispatch latency and field worker acceptance metrics.

2 Cadastral GIS Boundary Expansion

Ingest survey-grade shapefiles from Karnataka State Remote Sensing Applications Centre (KRSAC) for all 65 MCC wards and peripheral MUDA layouts.

3 Kannada Vernacular Voice & Text Input

Integrate speech-to-text models to allow Kannada-speaking citizens to lodge verbal complaints, automatically classified into the controlled municipal taxonomy.

TRANSPARENCY STATEMENT

Formal AI Tools Disclosure & Governance

AI Technologies Utilized in Development & Runtime:

1. Google Gemini API (gemini-1.5-flash):
Used strictly as assistive AI to suggest civic categories from description text.
AI NEVER makes jurisdiction, municipal authority, or dispatch decisions.
2. Antigravity AI Assistant:
Used for codebase architecture planning, test automation, and code review.

STATUTORY GOVERNANCE PRINCIPLE

100% Deterministic Civic Routing:

All civic routing, jurisdiction containment, and department dispatches are executed via deterministic PostGIS spatial SQL (ST_Covers).

Zero AI guessing or probabilistic classification is ever permitted for legal authority resolution.
All decisions are locked permanently to active jurisdiction version IDs in PostgreSQL.

PROJECT ACCESS & REPOSITORIES

Live Working MVP URL:

<https://adaptive-civic-routing.vercel.app>

GitHub Repository (master):

<https://github.com/kkaribasappakl-eng/adaptive-civic-routing>

Team Spark (HM26-E9C5):

HackMysuru 1.0 Phase 1 Submission